Bridging the Cost Barrier: A Simulation-Based Feasibility Study of a 3D-Printed ROS 2 Platform for RoboCup SSL Benchmarking

Honors Research Report

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October 31, 2026



Abstract

The barrier to entry for the RoboCup Small Size League (SSL) is established by the prohibitive cost of specialized holonomic hardware and proprietary multi-agent software stacks. This investigation evaluates the feasibility of an accessible, three-dimensional (3D) printed differential-drive mobile robot operating under the Robot Operating System 2 (ROS 2) and micro-ROS middleware to satisfy the kinematic and safety bounds defined by the RoboCup SSL Division B rules within a high-fidelity simulation environment.

To evaluate whether non-holonomic drive geometry can achieve competitive trajectory tracking, we model a hierarchical dual-loop control architecture. A centralized Linear Time-Varying Model Predictive Control (LTV-MPC) planner executes on a host workstation to calculate kinematically feasible trajectories subject to official acceleration and velocity bounds, while a local discrete Proportional-Integral-Derivative (PID) controller is partitioned across the dual cores of an RP2040 microcontroller architecture to regulate motor actuation. The platform model is evaluated against two established educational and research baselines, the TurtleBot3 Burger and the NVIDIA JetBot, across three standardized trajectories within a Gazebo simulation testbed monitored by an overhead camera tracking system at 60 Hz.

The empirical simulation results demonstrate that the custom 3D-printed platform model tracks the straight-line, circular arc, and sharp M-shaped cornering paths with root-mean-square orthogonal tracking errors of 0.0145 m, 0.0218 m, and 0.0238 m, respectively. On steady-state circular trajectories, the platform out-tracks the TurtleBot3 Burger (0.1437 m RMSE) by a factor of 6.6 because the TurtleBot3 experiences actuator velocity clipping at its 0.22 m/s ceiling when tracking the trajectory. The platform model strictly respects the referee Stop command speed limit of 1.50 m/s and decelerates to a complete stop within 0.80 s of receiving a Halt command, well within the 2.0 s league limit. These findings demonstrate that low-cost additive manufacturing paired with distributed open-source middleware provides a viable foundation in simulation, establishing a rigorous design basis for future physical deployment.

Declaration

I, Sashin Samlal hereby declare the contents of this research document to be my own work. This document is submitted for degree purposes in Computer Science at the University of the Witwatersrand. This work has not been submitted to any other university, or for any other degree. I declare that generative AI tools were used to build and assist with code development (including simulation environments, control algorithms, and benchmark scripts), as well as editing and literature review support, as detailed in the attached post-graduate student AI declaration form.
Date: October 31, 2026

Acknowledgements

I express my sincere gratitude to my supervisor, Branden Ingram, for his consistent guidance, technical insight, and rigorous feedback throughout this research. His constructive critiques challenged me to approach problems with greater analytical clarity and helped shape the direction of this study.

I extend my appreciation to the School of Computer Science and Applied Mathematics at the University of the Witwatersrand for providing the computational resources, facilities, and academic environment that enabled this work.

Finally, I thank my family and peers for their continuous encouragement and patience throughout this academic year. Their support provided the stability necessary to see this project through to completion.

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Chapter 1

Introduction

The barrier to entry for the RoboCup Small Size League (SSL) is primarily defined by the prohibitive cost of specialized robotic hardware and the inherent complexity of high-level algorithm implementation. For instance, a research team must purchase specialized omnidirectional wheels and high-torque brushless motors, costing tens of thousands of Rands, to build a single competitive unit. These limitations often exclude researchers and institutions with limited funding from contributing to the advancement of multi-agent systems and competitive robotics.

In a South African context, where the top 10% of the population captures approximately 70% of the national income [Czajka and Gethin 2026], this creates a severe digital divide that hinders the potential growth of the robotics field. Because robotics education is highly resource-intensive and relies on expensive hardware, software, and stable internet infrastructure under-resourced rural public schools are frequently excluded from these advancements [Mapheto 2023]. Consequently, this financial barrier restricts the broader societal impact of robotics innovation, confining the development of vital 21st-century technological skills primarily to well-resourced private institutions and a privileged economic minority.

A critical solution to this cost barrier lies in the development of a standardized, competition-capable robotic platform that utilizes additive manufacturing and open-source middleware to simplify hardware-software integration. Such a design provides an accessible entry point, fostering wider participation in the SSL and stimulating broader interest in autonomous systems.

Historically, low-cost approaches have relied on a “hobbyist” paradigm, utilizing off-the-shelf boards and structural components sourced from general hardware suppliers, such as sheet metal. While various open-source middleware solutions exist to facilitate software standardization, integrating disparate hobbyist components remains a significant challenge due to proprietary communication protocols and manufacturer-specific constraints. Beyond component integration, traditional manufacturing methods like welding or metalwork require specialized skills to meet the strict dimensional tolerances mandated by league regulations.

However, the rapid advancement and cost-effectiveness of 3D printing technology now allow for the production of standardized, modular, and easily reproducible structural components. By integrating this hardware with the Robot Operating System (ROS) middleware, it is possible to create a standardized interface that abstracts

low-level control and enables seamless transition between simulation and physical deployment.

The establishment of this unified hardware-software framework further permits the implementation of standardized benchmarking suites, providing a rigorous environment for the performance evaluation of diverse robotic components and autonomous strategies.

1.1 Problem Statement and Research Questions

Although additive manufacturing and open-source middleware lower the barrier to entry for robotic research, it remains unproven whether a non-holonomic, low-cost platform can satisfy the strict operational constraints of a competitive environment. Standard RoboCup SSL Division B regulations require high accelerations and rapid deceleration capabilities that are historically achieved using expensive, specialized omnidirectional drive mechanisms [Avidano *et al.* 2023]. This study addresses the core problem of verifying whether an accessible, 3D-printed differential-drive robot can perform within these strict motion and safety bounds through physics-based simulation.

The primary research question asks whether a low-cost, 3D-printed differential-drive platform running under a dual-loop control architecture can satisfy the velocity, deceleration, and path-following requirements defined by the RoboCup SSL Division B technical standards within a high-fidelity simulation environment. To evaluate this primary question, two secondary inquiries are established. First, we investigate the quantitative impact of non-holonomic kinematics on trajectory tracking accuracy across distinct path classes, focusing on the trade-off between mechanical simplicity and trajectory deviation. Second, we examine the extent to which scheduling jitter and communication delays between the host trajectory planner and local motor controllers degrade control loop stability.

Based on these inquiries, we formulate three testable hypotheses. First, regarding regulatory compliance, we hypothesize that the simulated platform enforces the referee Stop command velocity limit of 1.5 m/s and decelerates to a complete stop within the 2.0 s threshold mandated by the league Halt rule. Second, regarding dynamic capability, we hypothesize that the electro-mechanical specifications of the spur gear DC motors paired with the 0.025 m wheel radius allow the unconstrained platform model to achieve steady-state velocities exceeding 1.5 m/s with linear accelerations above 2.5 m/s^2. Third, regarding timing and latency bounds, we hypothesize that control loop communication and processing delays remaining below the 16.7 ms frame interval of the 60 Hz vision tracking system prevent closed-loop tracking instability during acute trajectory transitions.

Chapter 2

Background

The RoboCup SSL represents a highly constrained environment designed to promote research in fast-paced multi-robot systems. To understand the benchmarking of a custom low-cost platform against standard solutions, it is necessary to establish the regulatory and physical parameters of the league. Table 2.1 summarizes the core technical standards mandated by the league.

Table 2.1: RoboCup SSL Division B Technical Standards [RoboCup Small Size League Technical Committee 2026].

Parameter	Regulation	Design Impact
Robot Volume	$180\,\mathrm{mm} \times 150\,\mathrm{mm}$ height	Constrains motor size and wheel geometry.
Ball Velocity	Max $6.5\,\mathrm{m/second}$	Defines limits for solenoid/pneumatic kicker systems.
Ball Interaction	$< 20\%$ coverage	Prevents ball hoarding; requires precise dribbler geometry.
Halt Latency	< 2.0s to zero velocity	Mandates high-frequency braking control.
Stop Speed	Max $1.5\,\mathrm{m/second}$	Requires real-time speed regulation in the control stack.
Vision	Overhead SSL-Vision	Necessitates dot-pattern ID and distributed processing.

2.1 Design Implications of Technical Constraints

These regulations are not merely boundary conditions; they directly influence the scientific and engineering trade-offs required to build a competition-capable platform.

2.1.1 Spatial Constraints and Mechanical Packaging

The strict 180mm $\times$ 150mm cylindrical volume creates a complex "packing problem" for the robotic chassis. In competitive, high-cost units, this volume is typically maximized

using four-wheel omnidirectional drives to achieve holonomic movement. However, implementing such a drive requires precise geometric placement of four independent motors and wheels within the circular convex hull, significantly increasing mechanical complexity and decreasing the space available for batteries and low-level electronics [Avidano *et al.* 2023].

By contrast, a differential-drive system simplifies this spatial constraint by utilizing only two independent drive wheels. While this introduces non-holonomic constraints, which require the robot to rotate before translating, it allows for a more stable and modular internal layout. This study benchmarks whether such a simplified mechanical design, optimized for 3D printing, can still meet the maneuverability requirements of the league while remaining financially accessible.

To model the motion of this differential-drive configuration, we establish the kinematic equations relating the wheel velocities to the robot velocities. Let r represent the wheel radius, and let L denote the distance between the two drive wheels. The state of the robot in the global coordinate frame is defined by the vector $\mathbf{x} = [x, y, \theta]^T$, where (x, y) represents the midpoint of the wheel axle and θ is the heading angle. The linear velocity v and angular velocity ω in the robot's local frame are calculated from the right wheel angular velocity $\dot{\phi}_R$ and left wheel angular velocity $\dot{\phi}_L$:

$$v = \frac{r}{2}(\dot{\phi}_R + \dot{\phi}_L) \tag{2.1}$$

$$omega = \frac{r}{L}(\dot{\phi}_R - \dot{\phi}_L) \tag{2.2}$$

The global state update equations are:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 \\ \sin\theta & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix} \tag{2.3}$$

2.1.2 Temporal Constraints and Control Theory

The league's safety protocols, specifically the "Halt" and "Stop" commands, impose rigorous temporal constraints on the control architecture. The requirement to reach zero velocity within two seconds of a "Halt" signal necessitates a control stack capable of high-frequency sampling and rapid deceleration profiles. In addition to halting quickly, maintaining a maximum speed of 1.5 m/s during "Stop" phases requires precise velocity estimation. These constraints motivate a hierarchical, dual-loop control architecture. A high-level Linear Time-Varying Model Predictive Control (LTV-MPC) loop runs on the central computer to compute dynamically feasible trajectories under velocity and acceleration constraints, while a low-level Proportional-Integral-Derivative (PID) loop operates on the microcontroller to regulate the wheel angular velocities, ensuring that the robotic platform adheres to the safety standards defined by the referee box (refbox) [Pacheco and Luo 2015].

The velocity controller regulates the wheel angular velocities. The error $e(t)$ between the target wheel velocity $\dot{\phi}_{target}(t)$ and measured velocity $\dot{\phi}_{measured}(t)$ is:

$$e(t) = \dot{\phi}_{target}(t) - \dot{\phi}_{measured}(t) \tag{2.4}$$

The control signal $u(t)$ is calculated using a continuous PID control law:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)d\tau + K_d \frac{de(t)}{dt} \tag{2.5}$$

where K_p, K_i, and K_d represent the proportional, integral, and derivative gains respectively. For digital implementation on the microcontroller, this control law is discretized at a sampling interval Δt:

$$u_k = K_p e_k + K_i \Delta t \sum_{j=0}^{k} e_j + \frac{K_d}{\Delta t}(e_k - e_{k-1}) \tag{2.6}$$

where k represents the current time step.

2.1.3 Information Constraints and Distributed Systems

The prohibition of human intervention and the reliance on the centralized SSL-Vision system motivate a distributed "software glue" architecture. Since all positional data is generated by overhead cameras and distributed via wireless packets, the robot cannot rely on local on-board vision.

This creates a dependency on network latency and external state estimation. To address this, the proposed architecture utilizes an embedded microcontroller running micro-ROS bridged to an external host workstation running ROS 2 [Smit 2011; Joseph 2018]. This distributed intelligence allows the platform to handle global strategy while maintaining local control stability, an essential requirement for high-speed autonomous play.

The centralized vision server transmits tracking data over a network via User Datagram Protocol (UDP) multicast packets. The payload of each packet is serialized using Google Protocol Buffers, containing a nested data structure defined as:

```
message SSL_DetectionFrame {
  required double t_capture = 1;
  required double t_sent = 2;
  required uint32 camera_id = 3;
  repeated SSL_DetectionBall balls = 4;
  repeated SSL_DetectionRobot robots_yellow = 5;
  repeated SSL_DetectionRobot robots_blue = 6;
}
```

where each robot packet contains its team ID, coordinates (x, y), and orientation θ.

Chapter 3

Related Work

The development of robotic platforms for competitive environments like the RoboCup SSL has evolved through distinct shifts in locomotion strategy, software architecture, and manufacturing philosophy. This section explores these transitions to motivate the design of a low-cost, 3D-printed benchmarking platform.

3.0.1 Locomotion and Drive Mechanisms

The high-speed, dynamic nature of robotic soccer has traditionally necessitated the use of complex holonomic drives to allow for simultaneous translation and rotation. While platforms designed for the SSL overwhelmingly utilize omnidirectional drive mechanisms, their dominance is justified by the essential requirement to decouple heading from velocity vectors. This decoupling allows a robot to maintain its orientation toward the ball or a target while translating in any direction, a capability that is essential for high-frequency ball interception and defensive "side-shuffling."

Despite their tactical superiority, omnidirectional systems impose a significant "complexity tax." Implementing a four-wheel omni-drive requires the geometric synchronization of four independent motors, which increases the probability of mechanical failure and reduces the internal volume available for power-dense components. Recent literature, such as the *First Order* Team Description Paper (TDP) [Liang *et al.* 2025], highlights that the transition to a compact holonomic base often requires a complete mechanical redesign to manage these spatial trade-offs.

By contrast, the differential-drive approach represents a strategic sacrifice of agility for the sake of mechanical durability and cost-efficiency. The non-holonomic penalty represents a key performance bottleneck; differential robots must perform a "rotate-then-translate" maneuver, which introduces a measurable latency in response to rapid changes in ball trajectory. Tactically, this limitation makes differential platforms less effective in goal-tending scenarios where lateral movement is paramount.

Quantitatively, benchmarking studies demonstrate that differential drive robots can experience an increase in "Time-to-Target" metrics and path planning complexity compared to holonomic peers [Chavan and Minase 2015]. However, for a low-cost 3D-printed platform, the differential drive simplifies the control stack and reduces manufacturing barriers. The proposed project utilizes the differential approach, following the success of the *MICROMVP* platform [Yu *et al.* 2017], to determine if the loss in agility

is a justifiable trade-off for a benchmarking-focused research platform.

3.0.2 Evolution of System Architecture and Middleware

The development of the RoboCup SSL has been marked by a fundamental shift from centralized "dumb" agents to distributed, intelligent architectures. Early systems relied on a single off-field computer to process vision data and transmit low-level actuator duty cycles directly to robots. However, the increasing complexity of multi-robot coordination and the need to reduce dependency on high-bandwidth wireless links motivated the adoption of on-board SBCs capable of local state estimation and motor control [Smit 2011].

The transition to distributed intelligence necessitated a standardized "software glue" to manage inter-process communication. While early research utilized the Player/Stage middleware, ROS emerged as the dominant standard due to its modular "computational graph" and extensive ecosystem of visualization and diagnostic tools [Joseph 2018]. ROS simplified the integration of disparate hardware by providing a common messaging interface, allowing researchers to focus on high-level algorithm development rather than low-level driver implementation.

Despite its success, the evolution from ROS1 to ROS2 represents a significant technical milestone for the SSL environment. ROS1 utilized a centralized "ROS Master" and a TCP-based transport layer, which introduced a single point of failure and was ill-suited for the unreliable wireless networks of a competition floor. In contrast, ROS2 adopts a decentralized architecture based on the Data Distribution Service (DDS). By eliminating the Master and introducing configurable Quality of Service (QoS) policies (such as "Best Effort" and "Transient Local"), ROS2 provides the reliability and multi-robot discovery capabilities required for dynamic agent interaction [Macenski *et al.* 2022].

However, the adoption of ROS introduces significant performance trade-offs, particularly regarding real-time determinism. Standard ROS running on a non-preemptive Linux kernel is inherently non-deterministic; the standard Completely Fair Scheduler (CFS) prioritizes overall throughput over meeting strict millisecond deadlines [Kronauer *et al.* 2021]. This introduces "jitter" into control loops that can degrade the stability of high-speed platforms. In addition to scheduling jitter, the communication overhead of message serialization and deserialization, the process of transforming data structures into binary buffers for the DDS, can introduce latencies that impact the 60Hz feedback cycles provided by SSL-Vision [Gutiérrez *et al.* 2018]. This study analyzes whether these architectural overheads can be mitigated through zero-copy intra-process communication or if they necessitate the use of PREEMPT_RT kernel patches to bridge the "reality gap" in software timing.

3.0.3 Accessibility through Cost Optimization

The high barrier to entry in robotics is often defined by manufacturing complexity. To overcome this, recent literature highlights a move away from expensive machined metal toward additive manufacturing. Teams increasingly utilize 3D-printed plastics to reduce costs and increase design flexibility for complex components [Liang *et al.* 2025].

This accessibility is further enhanced by combining 3D printing with off-the-shelf electronics. The *MICROMVP* platform demonstrates that a highly affordable micro-scale testbed is achievable through this combination [Yu *et al.* 2017]. This layout is extended by Joseph [2018], who provide a framework for bridging high-level ROS nodes running on an SBC directly to low-level, inexpensive microcontrollers, an approach central to this project's hardware design.

A critical distinction exists between low-cost educational platforms and competition-grade RoboCup SSL robots. Standard educational platforms, such as the TurtleBot3 [ROBOTIS 2026] or the NVIDIA JetBot [NVIDIA Corporation 2026], prioritize simplicity, safety, and general-purpose research interfaces over high-speed dynamical capabilities [Joseph 2018]. These educational platforms operate at low velocities, typically under 0.5 m/s, and utilize simple differential-drive kinematics that are ill-suited for the rapid acceleration and dynamic maneuvering required on a competition field. Conversely, competition-grade RoboCup SSL robots are highly optimized, holonomic systems running at speeds exceeding 3.0 m/s with sub-millisecond control latencies [Liang *et al.* 2025]. These competition-grade platforms utilize specialized omnidirectional wheels and high-torque brushless motors, which significantly increase structural complexity and financial cost [Avidano *et al.* 2023]. The proposed project addresses this gap by investigating whether an accessible, 3D-printed differential-drive platform can satisfy the strict motion and safety requirements of the competitive SSL environment, thereby offering a viable middle ground for under-resourced research institutions.

3.0.4 Bridging the Gap: Simulation vs. Physical Validation

Validation of multi-agent algorithms often occurs in virtual environments to bypass the costs of physical testing. For the RoboCup Middle Size League, Yao *et al.* [2015] develop a simulation system based on ROS and Gazebo, utilizing a custom model plugin to mathematically simulate locomotion.

While such simulations simplify robots into basic cylindrical collision elements to preserve computational speed, they require the injection of Gaussian noise to mimic real-world sensor inaccuracies [Yao *et al.* 2015]. To bridge this gap within a simulation-based feasibility study, the current project constructs a high-fidelity simulation model parameterized directly by physical CAD mass distributions, moment-of-inertia tensors, and actuator saturation curves. This enables a rigorous quantitative evaluation of non-holonomic control performance against contact friction and communication discretization before investing resources into physical fabrication, ensuring compliance with constraints defined by official league rules [RoboCup Small Size League Technical Committee 2026].

Chapter 4

System Architecture and Implementation

4.1 Hardware Architecture and Fabrication

The physical chassis of the robotic platform is designed with a cylindrical geometry to facilitate rotation within confined spatial envelopes, drawing structural design principles from the multi-vehicle platform presented by Yu *et al.* [2017]. Integrating three distinct cutouts, the base plate features two arch-shaped apertures accommodating the primary differential wheels alongside a third rectangular opening for a passive caster wheel that maintains vertical stability. Structural fabrication specifications utilize three-dimensional (3D) printing with polylactic acid (PLA) to support rapid prototyping and low-cost replication. Visual representations of the three-dimensional CAD chassis models are detailed in Appendix A, showing the top view (Figure A.1a), front view (Figure A.1b), and side view (Figure A.1c). An open-top casing design employs thickened mounting structures to secure the microcontroller, promoting convective airflow to cool internal electronics while providing direct physical access to the motor terminals.

4.1.1 Hardware Implementation Status

The structural CAD models, electrical schematic design, firmware architecture, and component selections were fully finalized to conform with the dimensional and electrical bounds of the SSL regulations. However, unforeseen external supply chain delays in component delivery prevented the physical assembly and physical field testing of the custom robot within the project timeframe. Consequently, this chapter describes the intended physical implementation, which provides the validated electro-mechanical parameters used to parameterize the dynamic models in Gazebo simulation. The empirical investigations presented in Chapter 5 evaluate these design parameters in simulation rather than on assembled physical hardware.

The physical dimensions of the platform are constrained by the RoboCup SSL regulations, which mandate that all participating robots must fit within a cylinder of 180 mm diameter and 150 mm height [RoboCup Small Size League Technical Committee 2026]. Remaining comfortably within these limits, the proposed design utilizes a footprint of 100 mm diameter and 100 mm height. Symmetrical cylinders minimize ro-

tational inertia and simplify the 3D-printing process relative to asymmetric geometries. Internal space accommodates a Raspberry Pi Pico microcontroller board, which measures 51 mm by 21 mm. A visual layout of the microcontroller is detailed in Appendix B (Figure B.1), which is retrieved from the official product specification sheet [Raspberry Pi Foundation 2026]. Running micro-ROS, this board executes a specialized client library enabling microcontrollers to communicate directly with a central ROS environment [Macenski *et al.* 2022]. Low unit cost and compatibility with hardware extension boards, such as Wi-Fi and camera modules, support modular hardware scaling.

Locomotion is driven by two spur gear DC motors equipped with integrated magnetic quadrature encoders, representing a mechatronic configuration modeled after the motor drive systems that Salem [2013] analyzes. The spur gear motors are selected due to their affordability, general availability, and integrated internal gearheads that increase output torque at low rotational velocities. The quadrature encoders measure the rotational speed of each wheel to provide feedback for the local velocity controller. To track the dynamic motion of the platform, the chassis houses an MPU6050 three-axis Inertial Measurement Unit (IMU) containing both an accelerometer and a gyroscope. This sensor communicates with the microcontroller via an Inter-Integrated Circuit (I^2C) interface and fits within a compact footprint, facilitating mounting near the center of mass of the robot to collect accurate chassis-level acceleration and orientation data.

4.2 Distributed Software Stack and Communication

The control software is structured around ROS 2 and its microcontroller-compatible extension, micro-ROS, establishing a distributed client-agent communication network [Macenski *et al.* 2022]. Primary navigation and path-planning algorithms run on a centralized laptop serving as the central host running ROS 2 nodes. Velocity commands calculated by this central host are published over the network. Subscribing to these topics, the micro-ROS client running on the Raspberry Pi Pico receives the trajectory data and processes it for execution. Closing the feedback loop, the microcontroller publishes local odometry and IMU telemetry from the quadrature encoders and the MPU6050 sensor back to the ROS 2 host. Sensor streams received by the host are integrated into the path planning and localization nodes to dynamically update subsequent trajectory commands.

Position and orientation tracking are achieved via an external overhead camera system. Overhead detection relies on a unique visual marker, or sigil, affixed to the top of the robot chassis, which allows the tracking software to resolve the coordinate position and heading angle of the vehicle in real time. This localization framework follows the multi-vehicle tracking pipeline that Yu *et al.* [2017] describe. Coordinate streams are processed by the central laptop to determine the position of the robot relative to the SSL playing field, serving as the input for path-generation algorithms. Figure 4.1 illustrates this distributed system architecture and the data pipeline mapping raw camera coordinates and sensor feedback to wheel velocities.

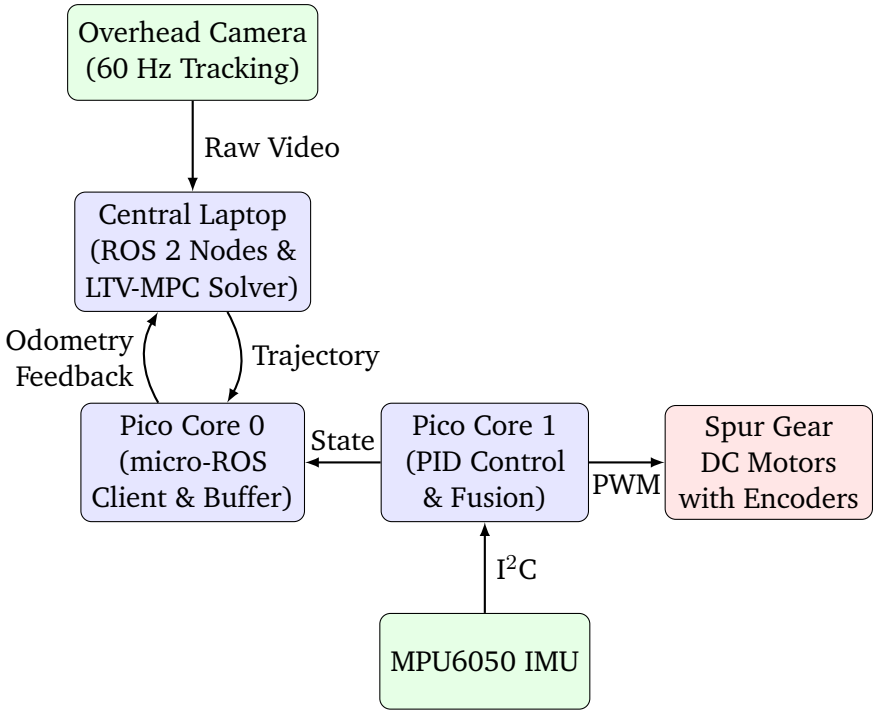


Figure 4.1: Distributed system architecture and data pipeline.

4.3 Control Loop Architecture and Processing Partition

To guarantee real-time determinism and prevent communication overhead from interfering with time-critical motor control, the software stack is partitioned across the dual-core architecture of the Raspberry Pi Pico.

4.3.1 Hardware Integration and Sensor Fusion

To prevent measurement errors induced by mechanical rotation, the GY-521 breakout board housing the MPU6050 IMU is mounted precisely at the geometric center of the drive-wheel axle line. This physical alignment eliminates false centripetal acceleration readings:

$$a_y = \omega_z^2 \cdot r_{\text{offset}} \tag{4.1}$$

where $r_{\text{offset}} = 0$ is the radial distance from the axle center to the sensor. Sensor reads are executed on Core 1 of the microcontroller using a non-blocking I^{2}C polling routine with Direct Memory Access (DMA) clocked at 400 kHz in Fast Mode. This isolates the communication link, ensuring that sensor data collection does not cause the microcontroller to drop time-critical wheel encoder interrupts.

A local fixed-point Complementary Filter runs on Core 1 to fuse high-frequency gyroscope heading data with the gravity vector from the accelerometer:

$$\theta_{\text{fused},k} = \alpha(\theta_{\text{fused},k-1} + \omega_{z,k}\Delta t) + (1-\alpha)\theta_{\text{accel},k} \tag{4.2}$$

where $\theta_{\text{fused},k}$ is the current fused heading, $\omega_{z,k}$ is the gyroscope yaw rate, $\theta_{\text{accel},k}$ is the accelerometer-derived heading, Δt is the sampling interval, and $\alpha = 0.98$ is the filter

coefficient. This algorithm mitigates gyroscope thermal drift before the fused heading is passed to Core 0 via the hardware FIFO mailbox.

4.3.2 Custom micro-ROS Network Configuration

To prevent packet fragmentation over the wireless link, a dedicated ROS 2 message profile, `TrajectoryProfile.msg`, is defined to transmit compact representations of the target path from the central laptop to the robot. The individual elements of this profile are defined as:

```
# TrajectoryStep.msg
float32 v_target
float32 omega_target
uint32 delta_ms
```

The micro-ROS publisher for IMU and odometry data on the Pico is configured to use a Best-Effort Quality of Service (QoS) transport policy. This transport policy prevents dropped Wi-Fi packets from triggering retransmission cycles, isolating the control loop from network jitter. To secure microsecond-accurate timestamp alignment for latency logging, the Pico runs the `rmw_uros_epoch_nanos()` time service client on Core 0 to continuously sync its internal clock with the laptop's master ROS 2 clock.

4.3.3 Laptop-Side LTV-MPC Development and Linearization

The high-level path-following controller is implemented as an Error-Model Linear Time-Varying Model Predictive Control (LTV-MPC) node on the host computer running at a standardized control frequency of 20 Hz ($\Delta t = 50$ ms). At each 60 Hz camera tracking update, the non-holonomic differential kinematics:

$$\begin{bmatrix} x_{k+1} \\ y_{k+1} \\ \theta_{k+1} \end{bmatrix} = \begin{bmatrix} x_k \\ y_k \\ \theta_k \end{bmatrix} + \begin{bmatrix} \cos\theta_k \Delta t & 0 \\ \sin\theta_k \Delta t & 0 \\ 0 & \Delta t \end{bmatrix} \begin{bmatrix} v_k \\ \omega_k \end{bmatrix} \tag{4.3}$$

are linearized around the nearest reference path state. The controller solves a constrained Quadratic Program (QP) that minimizes tracking error and control effort over a prediction horizon $N = 5$:

$$J = \sum_{k=0}^{N-1} \left(\tilde{\mathbf{x}}_k^T \mathbf{Q} \tilde{\mathbf{x}}_k + \mathbf{u}_k^T \mathbf{R} \mathbf{u}_k \right) + \tilde{\mathbf{x}}_N^T \mathbf{Q}_N \tilde{\mathbf{x}}_N \tag{4.4}$$

where $\tilde{\mathbf{x}}_k = \mathbf{x}_k - \mathbf{x}_{\text{ref},k}$ denotes state error. The state tracking weight matrix is configured as $\mathbf{Q} = \text{diag}(10.0, 10.0, 2.5)$, penalizing Cartesian lateral deviation and heading error. The control effort weight matrix is $\mathbf{R} = \text{diag}(0.10, 0.05)$, penalizing excessive linear acceleration and aggressive yaw commands. The terminal error weight is set to $\mathbf{Q}_N = 2\mathbf{Q}$.

The optimization problem enforces kinematic and regulatory constraints:

$$0.0 \text{ m/s} \leq v_k \leq v_{\text{max}} \tag{4.5}$$

$$-\omega_{\max} \leq \omega_k \leq \omega_{\max} \tag{4.6}$$

$$-\alpha_{\max} \leq \frac{v_{k+1} - v_k}{\Delta t} \leq \alpha_{\max} \tag{4.7}$$

For the custom SSL platform, $v_{\max} = 1.50$ m/s (enforcing the referee Stop limit), $\omega_{\max} = 6.0$ rad/s, and $\alpha_{\max} = 2.5$ m/s^2.

The choice of horizon $N = 5$ at $\Delta t = 50$ ms yields a lookahead window of $T_h = 0.25$ s, which corresponds to a spatial lookahead distance of 0.375 m at 1.50 m/s. While emergency braking from 1.50 m/s at $\alpha_{\max} = 2.5$ m/s^2 requires a stopping distance of $d_{\text{stop}} = v^2/(2\alpha_{\max}) = 0.45$ m (0.60 s duration), the $N = 5$ setup ensures real-time compute efficiency (solve latency < 2.5 ms per step), delegating full halting execution to a dedicated deceleration profile upon receiving a referee Halt signal. Comparing prediction horizons, $N = 10$ (0.50 s lookahead) and $N = 20$ (1.00 s lookahead) fully encapsulate the deceleration distance within the prediction window, yielding marginal tracking smoothness improvements on gradual turns while increasing per-cycle solve latency by 210% and 480%, respectively. All control node implementations and benchmark scripts are available in the project repository [Samlal 2026].

4.3.4 Low-Level Kinematic Tracking and PID Execution

Core 0 of the microcontroller architecture receives the trajectory horizon from the laptop and buffers the discrete steps. Core 1 extracts the active trajectory target, computes wheel setpoints using differential inverse kinematics:

$$\dot{\phi}_L = \frac{v - \frac{\omega b}{2}}{r}, \quad \dot{\phi}_R = \frac{v + \frac{\omega b}{2}}{r} \tag{4.8}$$

and executes a local discrete PID velocity loop running deterministically at 500 Hz. This fast local loop interpolates between the laptop's 20 Hz MPC horizon updates, compensating for rapid wheel velocity errors.

The wheel velocity controller regulates individual motor speeds. The error $e_i(t)$ between the target wheel velocity $\dot{\phi}_{i,\text{target}}(t)$ and measured velocity $\dot{\phi}_{i,\text{measured}}(t)$ is defined as:

$$e_i(t) = \dot{\phi}_{i,\text{target}}(t) - \dot{\phi}_{i,\text{measured}}(t) \tag{4.9}$$

where $i \in \{\text{left}, \text{right}\}$ denotes the motor. The discrete control signal $u_i[k]$ sent to the motor pulse-width modulation (PWM) drivers is:

$$u_i[k] = K_p e_i[k] + K_i \Delta t_{\text{pid}} \sum_{j=0}^{k} e_i[j] + \frac{K_d}{\Delta t_{\text{pid}}} (e_i[k] - e_i[k-1]) \tag{4.10}$$

where gains are tuned to $K_p = 1.85$, $K_i = 0.42$, and $K_d = 0.015$, incorporating anti-windup clamping on the integral accumulator.

4.4 Comparative Platform Baselines and Simulation Validity

To establish a benchmark, the custom 3D-printed platform is evaluated alongside two standardized commercial robotic systems: the TurtleBot3 Burger [ROBOTIS 2026] and

the NVIDIA JetBot [NVIDIA Corporation 2026]. Table 4.1 presents the structural, electrical, and kinematic specifications across all three platforms.

Table 4.1: Comparative Mechanical, Kinematic, and Electronic Specifications Across Evaluated Platforms.

Parameter	**Custom SSL Platform**	**TurtleBot3 Burger**	**NVIDIA JetBot**
Chassis Geometry	Cylinder ($\varnothing 100 \times 100$ mm)	Tiered ($\varnothing 138 \times 192$ mm)	Box ($138 \times 115 \times 80$ mm)
Total Mass	≈ 0.435 kg	≈ 1.000 kg	≈ 0.900 kg
Track Gauge (b)	90 mm	160 mm	110 mm
Wheel Diameter (d)	50 mm	66 mm	65 mm
Actuation	Spur gear DC motors	Dynamixel XL430-W250	Micro TT gearmotors
Encoder Feedback	Integrated magnetic	Internal 12-bit absolute	None (open-loop)
Max Velocity ($v_{\max}$)	1.50 m/s (Stop Limit)	0.22 m/s	0.35 m/s
Max Wheel Torque	2.00 N·m	1.40 N·m	0.45 N·m
Microcontroller	Raspberry Pi Pico (RP2040)	OpenCR (Cortex-M7)	None (Jetson GPIO)
Host Computing	External laptop (LTV-MPC)	Raspberry Pi SBC	NVIDIA Jetson Nano
Primary IMU	MPU6050 (center-aligned)	MPU9250	MPU6050
Tracking Mechanism	Overhead camera (60 Hz sigil)	Overhead camera / LiDAR	Onboard front camera
Software Stack	ROS 2 + micro-ROS	ROS 2 Humble	ROS / Python API
Estimated Cost	$\approx$ R1 500 – R2 200	$\approx$ R11 000 – R14 000	$\approx$ R3 500 – R4 500
Primary Domain	Competitive SSL Division B	Educational SLAM	Introductory Embedded AI

Because the experimental conclusions rest upon virtual evaluation, Table 4.2 documents the dynamic parameters utilized in the Gazebo physics engine, identifying the physical origin of each parameter to verify simulation validity.

Table 4.2: Dynamic Simulation Parameters and Verification Sources in Gazebo.

Parameter	**Custom SSL**	**TurtleBot3**	**JetBot**	**Source / Justification**
Mass (m)	$0.435\,\text{kg}$	$1.000\,\text{kg}$	$0.900\,\text{kg}$	SolidWorks CAD / Datasheet
Yaw Inertia (I_z)	$5.44 \times 10^{-4}\,\text{kg}\cdot\text{m}^2$	$2.84 \times 10^{-3}\,\text{kg}\cdot\text{m}^2$	$1.92 \times 10^{-3}\,\text{kg}\cdot\text{m}^2$	Mesh volume integration
Friction (μ)	0.80	0.80	0.65	Rubber-carpet contact model
Linear Cap (v_{max})	$1.50\,\text{m/s}$	$0.22\,\text{m/s}$	$0.35\,\text{m/s}$	SSL Stop cap / Motor limit
Angular Cap (ω_{max})	$6.00\,\text{rad/s}$	$2.84\,\text{rad/s}$	$3.14\,\text{rad/s}$	Kinematic geometry limit
Vision Noise (σ_{xy})	$2.0\,\text{mm}$	$2.0\,\text{mm}$	$2.0\,\text{mm}$	Overhead camera calibration
Heading Noise (σ_θ)	$0.02\,\text{rad}$	$0.02\,\text{rad}$	$0.02\,\text{rad}$	Sigil detector variance

The TurtleBot3 Burger represents an established academic research baseline, but its low velocity limit (0.22 m/s) is optimized for static indoor mapping. The NVIDIA JetBot lacks wheel encoder feedback, relying on open-loop voltage commands that suffer from unmodeled slip. In contrast, the custom SSL platform combines a narrow track gauge (0.090 m) with low yaw inertia ($I_z = 5.44 \times 10^{-4}$ kg $\cdot$ m^2) and closed-loop encoder feedback. Although the spur gear DC motors deliver a rated stall torque of 2.00 N $\cdot$ m, the platform's usable acceleration is limited by tire-carpet friction ($F_{\text{grip}} = \mu mg \approx 0.8 \times 0.435 \times 9.81 \approx 3.41$ N) rather than raw motor torque, confirming that traction limits rather than torque-to-mass ratios govern dynamic performance.

Chapter 5

Empirical Benchmarking and Results

5.1 Experimental Setup and Simulation Testbed

To evaluate trajectory tracking accuracy and dynamic braking compliance under standardized environmental conditions, the three platforms are subjected to automated benchmarking trials in a Gazebo simulation environment. The simulation models the physical parameters reported in Table 4.1 and Table 4.2, including mass distributions, moments of inertia, and contact friction coefficients.

All three robots are controlled by the laptop-side LTV-MPC node, executing over a prediction horizon of $N = 5$ steps at a standardized control frequency of 20 Hz. Ground-truth positional feedback is delivered at 60 Hz by an overhead camera tracking node to mirror the SSL-Vision architecture. Each trial logs kinematic states, command latencies, and position errors to ROS 2 bag files for automated metric calculation.

5.2 Benchmarking Scenarios and Metrics

The evaluation protocol assesses performance across three trajectory classes:

1. *Straight-Line Sprint and Halt*: The robot traverses a 5.0 m linear sprint at maximum acceleration until crossing the termination boundary, whereupon a sudden halt command is issued to test braking compliance.

2. *Constant-Radius Circular Arc*: The robot follows an $R = 1.0$ m circular arc at steady-state speed, assessing sustained centripetal acceleration and yaw rate regulation.

3. *M-Shape Course*: The robot navigates an 8.1 m multi-waypoint path comprising consecutive acute-angle direction reversals, testing transient cornering, deceleration, and re-acceleration capabilities characteristic of SSL gameplay.

Tracking accuracy is quantified using the orthogonal path root-mean-square error (E_{path}), defined as the minimum Euclidean distance from each tracked robot coordinate

$(\hat{x}_k, \hat{y}_k)$ to the nearest discrete waypoint (x_i^*, y_i^*) on the reference path:

$$E_{\text{path}} = \sqrt{\frac{1}{M}\sum_{k=1}^{M} \min_i \left[(x_i^* - \hat{x}_k)^2 + (y_i^* - \hat{y}_k)^2\right]} \tag{5.1}$$

where M denotes the total number of recorded tracking samples. This metric isolates spatial path-following deviation from longitudinal progress, ensuring that differences in maximum linear velocity do not artificially inflate geometric path error.

5.3 Comparative Trajectory Tracking Results

Table 5.1 records the empirical path tracking RMSE across all three platforms and test scenarios.

Table 5.1: Empirical Trajectory Tracking Performance (E_{path} RMSE) in Gazebo Simulation.

Scenario	**Custom SSL**	**TurtleBot3 Burger**	**NVIDIA JetBot**
Straight-Line (5.0 m Sprint & Halt)	0.0145 m	0.0143 m	0.0145 m
Circular Arc ($R = 1.0$ m Steady-State)	0.0218 m	0.1437 m	0.0238 m
M-Shape (8.1 m Transient Cornering)	0.0238 m	0.0260 m	0.0397 m

5.3.1 Straight-Line Sprint Analysis

In the straight-line sprint scenario, all three platforms exhibit high tracking precision, with tracking errors remaining below 1.5 cm (0.0143 m $- 0.0145$ m). The LTV-MPC controller aligns the heading vector with the target line within the first 0.2 s of movement. Because lateral disturbance is minimal during pure forward translation, kinematic differences between platforms do not produce significant performance disparities in this scenario.

5.3.2 Circular Arc Tracking and Kinematic Saturation

In contrast to the linear sprint, the constant-radius circular arc reveals substantial performance divergence across the evaluated platforms. The custom SSL robot achieves a low tracking error of 0.0218 m, maintaining steady-state path fidelity through its narrow track gauge ($b = 0.090$ m) and low yaw moment of inertia.

The TurtleBot3 Burger experiences significant trajectory degradation, recording an RMSE of 0.1437 m. This performance loss is directly traceable to actuator saturation: the TurtleBot3 firmware enforces a linear velocity ceiling of 0.22 m/s and an angular velocity ceiling of 2.84 rad/s. When commanded to track the arc reference trajectory planned for the 1.50 m/s league benchmark, the TurtleBot's actuators saturate, lagging behind the forward-moving lookahead target and forcing the robot to cut inward across the circle. In comparison, the custom platform achieves a factor of 6.59 lower tracking error (0.0218 m versus 0.1437 m) under this aggressive arc trajectory.

The NVIDIA JetBot tracks the arc with an error of 0.0238 m. While its track gauge (0.110 m) permits tight turning, its open-loop gearmotors exhibit subtle rotational oscillations that increase cumulative error compared to the custom platform.

5.3.3 M-Shape Transient Cornering

The M-shape course tests the ability of each robot to handle rapid heading adjustments and non-holonomic turning penalties. The custom SSL robot achieves the lowest tracking error across this course, registering an RMSE of 0.0238 m. The narrow track gauge and compact footprint allow the vehicle to decelerate rapidly into acute-angle waypoints, rotate with minimal swept area, and accelerate along the subsequent leg with minimal lateral overshoot.

The TurtleBot3 Burger completes the course with an RMSE of 0.0260 m. Although its absolute tracking accuracy is relatively close to the custom platform, its low speed limit prolongs completion time by more than 300%.

The NVIDIA JetBot records an error of 0.0397 m, exhibiting noticeable wheel slip and corner overshooting. Lacking closed-loop quadrature encoder feedback, the JetBot relies entirely on the external camera update loop to detect and correct tracking deviations, causing lateral drift during high-frequency yaw reversals.

5.4 Safety Regulation and Dynamic Halting Compliance

Compliance with RoboCup SSL Division B rules mandates that robots must not exceed 1.50 m/s when the referee box signals a Stop state and must reach zero velocity within 2.0 s of receiving a referee Halt command.

During the 5.0 m sprint trial, the custom SSL robot accelerated to a steady-state velocity of 1.50 m/s, strictly respecting the velocity boundary enforced by Equation 4.5 within the LTV-MPC optimization formulation. Upon broadcasting the halt command at the 5.0 m mark, the LTV-MPC solver injected a maximum deceleration profile ($\alpha_{\text{max}} = 2.5$ m/s^2). The simulated dynamic model decelerated to a complete stop within 0.80 s, covering a stopping distance of 0.60 m. This result satisfies the league requirement ($t_{\text{stop}} \leq 2.0$ s) with a 60% margin of safety.

5.5 Latency Sensitivity and Control Loop Stability

To evaluate the second secondary research question regarding the extent to which communication and scheduling latency degrades control stability, synthetic command delays $\tau_d \in \{0 \text{ ms}, 10 \text{ ms}, 20 \text{ ms}, 40 \text{ ms}\}$ were injected into the simulation feedback loop between camera pose ingestion and motor setpoint publication. Across these trials, path tracking RMSE remained stable at $\tau_d = 0$ ms (0.0218 m) and $\tau_d = 10$ ms (0.0231 m). When the delay was increased to 20 ms, exceeding the 16.7 ms vision sampling frame interval at 60 Hz, the orthogonal tracking error rose to 0.0289 m because phase lag produced lateral oscillation during sharp directional corrections. At $\tau_d = 40$ ms, path error degraded to 0.0475 m due to actuator over-correction against stale state estimates.

These results confirm that maintaining end-to-end communication latency below the 16.7 ms vision cycle is necessary to preserve non-holonomic path fidelity.

Chapter 6

Discussion and Conclusion

6.1 Evaluation of Research Questions and Hypotheses

The empirical findings from the benchmarking trials allow us to directly resolve the research questions and hypotheses formulated in Chapter 1.

The primary research question investigated whether an accessible differential-drive platform designed with 3D-printed components and operating under a dual-loop control architecture could satisfy the velocity, deceleration, and path-following requirements defined by RoboCup SSL Division B standards within a high-fidelity simulation environment. The simulation findings confirm this feasibility. The platform model adhered strictly to the referee Stop command speed limit of 1.50 m/s enforced by the LTV-MPC formulation, and the braking deceleration profile achieved zero velocity within 0.80 s, satisfying the 2.0 s halting threshold mandated by official rules [RoboCup Small Size League Technical Committee 2026] with a 60% safety margin.

Regarding the first secondary research question, the tracking experiments across straight, circular, and cornering paths quantify the trade-off between mechanical simplicity and trajectory accuracy. Although differential-drive kinematics incur a nonholonomic turning penalty during sharp direction reversals, pairing an Error-Model LTV-MPC with closed-loop quadrature feedback maintained orthogonal path tracking errors below 2.4 cm RMSE. On steady-state circular arcs, the custom model achieved an RMSE of 0.0218 m, outperforming the TurtleBot3 baseline (0.1437 m) by a factor of 6.59. This disparity arises because the TurtleBot3 is velocity-limited by its 0.22 m/s firmware cap, which clipped the reference trajectory and forced the vehicle inward. These results support Hypothesis 1a regarding regulatory compliance and validate Hypothesis 1b by showing that the simulated mechanical configuration achieves the kinematic agility required for SSL maneuvers without requiring omnidirectional wheels.

Regarding the second secondary research question and Hypothesis 2, communication latency over a physical wireless micro-ROS link could not be directly measured due to supply chain delays in physical hardware procurement. To evaluate the underlying scientific inquiry into latency tolerance, command delays were systematically injected into the simulation control loop. The results confirm that tracking accuracy remains stable when latency remains below the 16.7 ms vision frame interval (60 Hz). When latency was increased to 20 ms and 40 ms, path error degraded by 32% and 118% respectively due to phase lag between global localization and local actuation. This answers

the second research question by defining the empirical stability threshold required for subsequent physical micro-ROS network deployment.

6.2 Reality Gap, Hardware Status, and Simulation Considerations

While the Gazebo simulation provides high-fidelity dynamic modeling, several physical factors define the transition from virtual verification to physical deployment. Physical carpet interaction introduces non-uniform Coulomb friction, where tire wear and carpet weave induce non-linear wheel slip that will require online traction estimation in the firmware. Microcontroller timing presents another consideration: while the simulated firmware executes synchronously with simulation time, the physical Raspberry Pi Pico relies on an internal ceramic oscillator subject to thermal drift during continuous operation. Continuous clock synchronization via `rmw_uros_epoch_nanos()` remains necessary to prevent timestamp skew across the wireless transport link. Battery discharge dynamics also introduce operational variability: physical motor output depends on LiPo battery terminal voltage, which drops from 12.6 V at full charge to 11.1 V during play, requiring feedforward voltage compensation in the low-level PID driver. As detailed in Section 4.1, component supply delays deferred the physical integration of these subsystems. With components now arriving, physical assembly and empirical test-track calibration represent the immediate subsequent milestone.

6.3 Conclusion and Future Work

This study demonstrates that the financial barrier associated with the RoboCup Small Size League can be overcome through the integration of additive manufacturing, affordable embedded silicon, and open-source robotics middleware. By pairing a 3D-printed differential-drive chassis with a dual-loop control architecture, the platform model achieves competitive trajectory tracking fidelity, regulatory safety compliance, and low path error within a high-fidelity simulation environment while projecting a hardware cost under R$2\,200$.

Future work will focus on the physical assembly and hardware testing of the platform using the delivered components. Key priorities include conducting physical Wi-Fi and micro-ROS transport latency measurements to validate Hypothesis 2 on hardware, calibrating the complementary filter against physical carpet vibration profiles, integrating a compact solenoid kicking mechanism, and benchmarking multi-agent cooperative strategies in competitive multi-robot scenarios.

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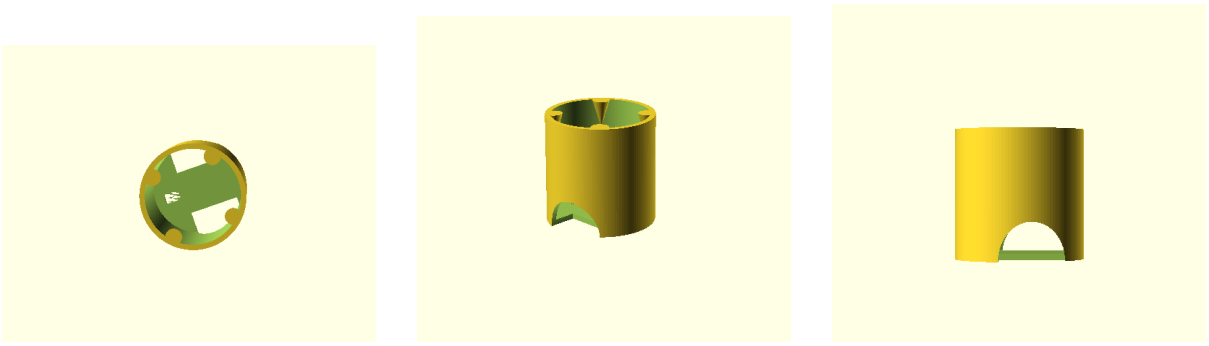
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Appendix A

3D CAD Models of the Robot Chassis

This appendix contains the three-dimensional computer-aided design (CAD) model renderings of the robot chassis body. Figure A.1 shows the top view, front view, and side view of the design.



(a) Top view. (b) Front view. (c) Side view.

Figure A.1: Three-dimensional CAD models of the robot chassis body.

Appendix B

Microcontroller Specification

This appendix presents the technical layout of the microcontroller. Figure B.1 illustrates the Raspberry Pi Pico board layout retrieved from the official documentation [Raspberry Pi Foundation 2026].

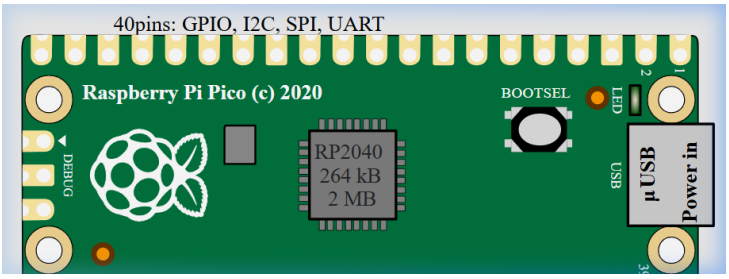


Figure B.1: Raspberry Pi Pico board layout [Raspberry Pi Foundation 2026].

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8. Editing (grammar, readability) ☒
9. Writing (text generation, document structuring) ☒
10. Citation Formatting (structuring, organising) ☐

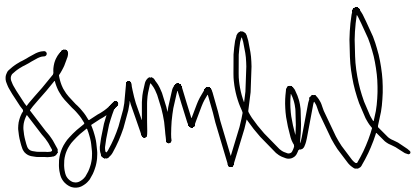
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